

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2016

BIOLOGY

9648/03

Applications Paper and Planning Question

26 Aug 2016

PAPER 3

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
TOTAL	72

This document consists of **16** printed pages.

1. The first restriction enzyme was discovered in *Escherichia coli* in the 1960s, and thousands more have been discovered since. These enzymes opened the path to powerful research tools which scientists use to sequence genomes through the process of genetic engineering.

(a) Compare the roles of restriction enzymes occurring naturally in bacteria and those used in genetic engineering. [2]

1. **Similarity 1** – Both cut dsDNA at specific sites which have complementary 3D conformation to the restriction enzyme;
2. **Similarity 2** – Cut phosphodiester bonds between adjacent nucleotides;
3. **Difference** – Restriction enzymes found naturally in bacteria cut/cleave/hydrolyse foreign/viral DNA to destroy them while those used in genetic engineering cut/cleave/hydrolyse vectors/gene of interest to allow gene of interest to be inserted into the vector;

Note: students need to have 1 similarity and 1 difference to score full marks

(b) A gene, *URA3*, was discovered to be a potential marker gene for genetic engineering. This gene is obtained from yeast and codes for orotidine 5'-phosphate decarboxylase, which converts 5-fluoroorotic acid (5-FOA) into the toxic compound 5-fluorouracil, so any cells carrying the *URA3* gene will not survive in the presence of 5-FOA.

In an investigation of the effectiveness of *URA3* as a marker gene, researchers created a plasmid vector shown in Fig. 1.1.

A eukaryotic gene was inserted into this plasmid and the mixture was added to competent *Escherichia coli* for transformation to take place. These bacteria were then grown on agar plates containing X-gal and 5-FOA.

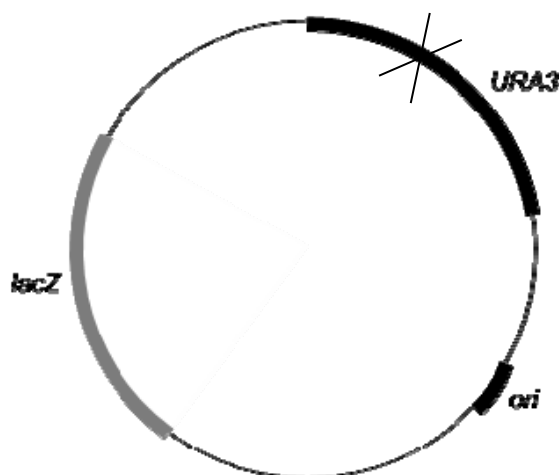


Fig. 1.1

(i) The researchers observed that the recombinant bacteria appeared as blue colonies on agar plates containing X-gal and 5-FOA. Using this information, draw a cross (X) on the plasmid in Fig. 1.1 to indicate where the eukaryotic gene was inserted. [1]

The cross should be within the *URA3* gene.

(ii) Account for the observations stated in b(i). [4]

1. **The *URA3* gene will be disrupted because the gene of interest has been inserted into the plasmid within the *URA3* gene;**

2. **5-FOA will not be converted into the toxic compound 5-fluorouracil, and so there will be growth of bacteria colonies on the agar plates;**
3. **Bacteria with the recombinant plasmid will express a *lacZ* gene/*lacZ* gene is intact;**
4. **X gal will be hydrolysed to form blue compound → colonies appear blue;**

(iii) State the purpose of the *lacZ* gene. [1]

Select for transformed bacteria (and not untransformed bacteria);

(iv) Explain why the bacteria containing the recombinant plasmid should be further subjected to gene probing. [3]

1. **To confirm the presence of the eukaryotic gene in the plasmid;**
 2. **A gene probe, which is single-stranded radioactive DNA complementary to the gene of interest → will specifically bind to the (denatured) gene of interest;**
 3. **Bacteria colonies with the gene of interest will expose photographic film (owtte);**
- OR**
1. **DNA other than the gene of interest could have been inserted into the vector/(for marking only: mutation could have occurred in the URA3 gene);**
 2. **A gene probe, which is single-stranded radioactive DNA complementary to the gene of interest → will not bind to any DNA segment in the bacteria (which has been denatured);**
 3. **Bacteria colonies will therefore not expose the photographic film (owtte);**

(c) The differences between the formation of genomic and cDNA libraries are due to the original source of genetic material used. In genomic libraries, the genetic material used is the genome of a haploid cell while in cDNA libraries, mRNA from a specialised cell is extracted.

(i) Describe two differences in the processes involved in the formation of these libraries. [2]

Basis of comparison	Genomic library	cDNA library
Addition of DNA linkers	1. No	Yes;
Use of DNA ligase	2. Once – formation of recombinant vector/DNA	Twice – Add linkers to ds cDNA and formation of recombinant vector/DNA;
Reverse transcription/ Use of enzymes involved in reverse transcription	3. No	Yes – reverse transcriptase, RNase H, DNA Pol;

(ii) State two applications of cDNA libraries. [2]

1. **To study gene expression;**
2. **To locate a gene of interest to be inserted into a cloning vector;**
3. **To use as a gene probe;**
4. **To use as a primer for PCR;**

[Total: 15m]

2. Gel electrophoresis is a technique that is widely used in molecular biology to separate DNA fragments based on molecular mass. The molecular mass of DNA fragments may differ due to the number of nucleotides present in the DNA fragment as well as chemical modifications made to the DNA.

Fig. 2.1 shows the gel electrophoregram of the DNA fragments taken from three sources. DNA fragments from all three samples are known to have equal lengths.

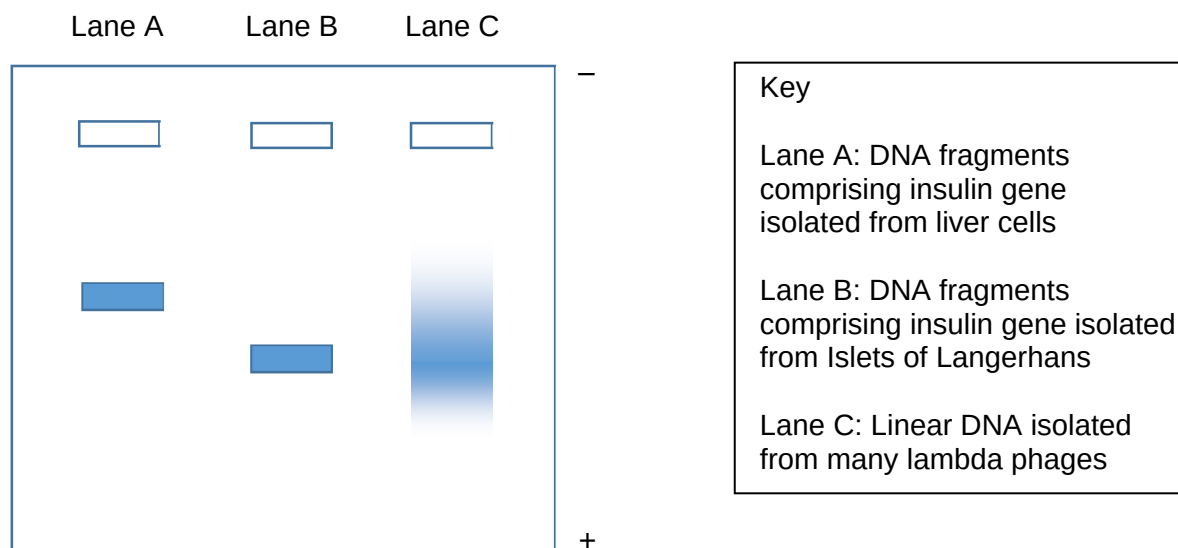


Fig. 2.1

- (a) Describe the role of the buffer solution in gel electrophoresis. [2]

1. To maintain the pH;
2. So that charge on DNA will not change, affecting the migration rate of DNA;
3. To conduct electricity across the gel by providing ions (for the separation of DNA fragments);

- (b) (i) Describe and explain the position of the band in lane A with respect to the band in lane B. [3]

1. DNA fragment migrated shorter distance than in lane B/ to a position of higher kilo basepair than in lane B/vice versa; (compulsory point to obtain full marks)
2. DNA fragment that contain insulin gene from the liver cell may be highly methylated ;
3. as it is not expressed in liver cell / found in heterochromatin region / region of transcriptionally inert DNA;
4. hence it slows down the movement of the DNA fragment due to the size of the methyl groups;

- (ii) Fig. 2.2 shows the DNA of a lambda phage.

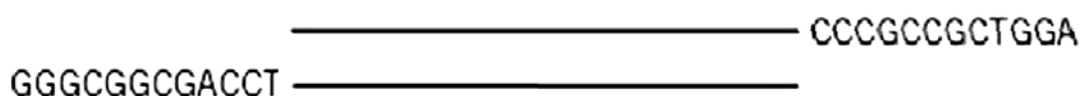


Fig 2.2

Describe and explain the band pattern of the lambda phage DNA in lane C. [4]

1. *Lambda DNA resulted in 1 smear / owtte; (compulsory point to obtain full marks)
2. DNA of lambda phage has (long) sticky ends;
3. Hence they can anneal with other lambda DNA by complementary base pairing;
4. Ref to multiple annealing can produce fragments of many different sizes/lengths (resulting in the smear)/ Annealing of complementary sticky ends at both strands, forming circular DNA;
5. Small size of circular DNA migrated faster (through the pores of the agarose gel);

- (c) Cyclopselemia is a new genetic disease recently discovered in a small town called X-mansion. A patient suffering from cyclopselemia has red eyes and cannot control the amount of light entering his eyes. They have to constantly wear sun glasses. Scientists used information from a pedigree chart as well as RFLP analyses of the individuals in the pedigree chart to determine the allele that is responsible for this genetic disease.

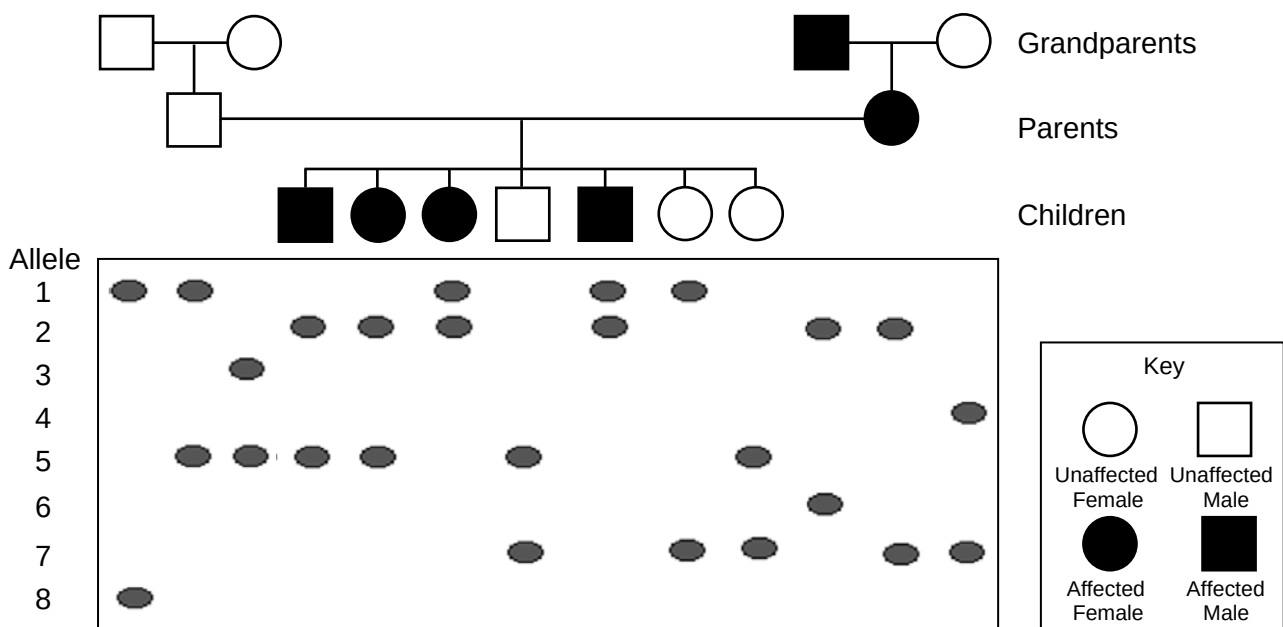


Fig. 2.3

Outline a process that can be used to determine the presence of cyclopselemia in a foetus. [5]

1. Isolate DNA from foetus and amplify via PCR;
2. Subject to restriction digest with the same restriction enzyme used in the RFLP analysis above;
3. Separate restriction fragment via gel electrophoresis;
4. Ref to southern blotting / Transfer fragment to nitrocellulose membrane;
5. Single-stranded radioactive probe binds/anneals to region of interest (via complementary base pairing);
6. Bands are visualised via autoradiography;
7. *Presence of one RFLP allele 2 indicates presence of disease; (compulsory point to obtain full marks)

[Total: 14m]

3. Embryonic stem cells, which are derived from the inner cell mass of mammalian blastocysts, are used in research to understand disease development and develop future cell-based therapies for currently untreatable diseases.

In 2006, a team of scientists led by Shinya Yamanaka discovered that differentiated cells can be isolated and modified to an embryonic-like state. These cells, subsequently designated as induced pluripotent stem cells (iPSCs), have been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine. Fig. 3.1 shows a potential application of human iPSCs in gene therapy.

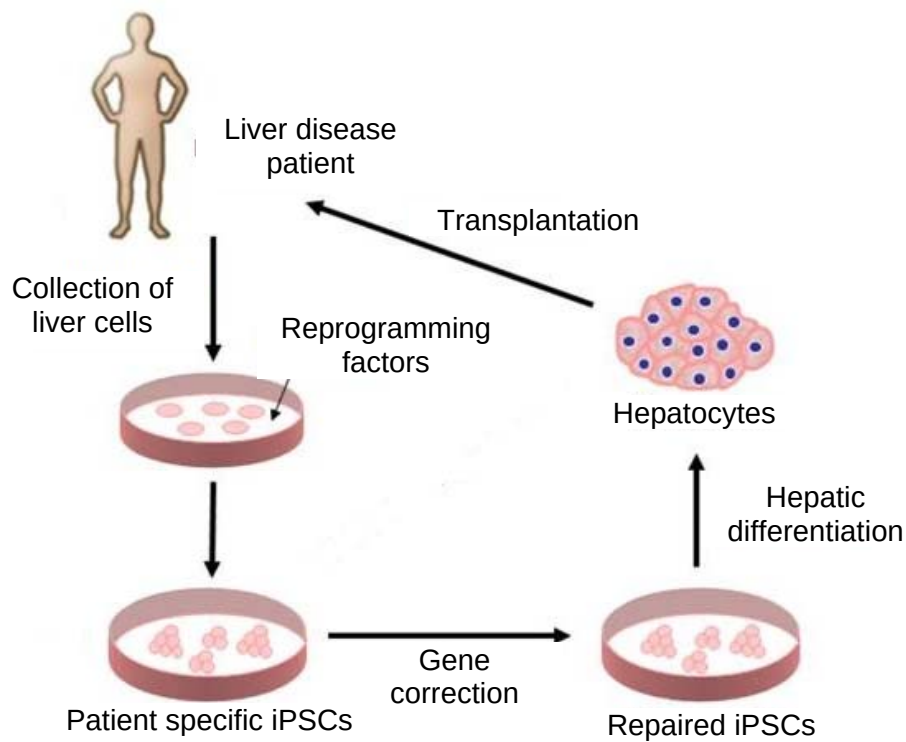


Fig. 3.1

(a) (i) State two similarities between the features of embryonic stem cells and iPSCs. [2]

1. Both are pluripotent (and can be multipotent, but not totipotent);
2. Both have the potential to become any cell type in the adult body but not those of the extra-embryonic membranes;
R! Only stating 'both can differentiate' without elaboration
3. Both have self-renewing capabilities / active telomerase / can divide continually via mitosis;
4. Both are unspecialised/undifferentiated;
5. Both have similar DNA methylation patterns;

(ii) State and explain two advantages of using iPSCs over embryonic stem cells in gene therapy. [4]

1. iPSCs can be obtained directly from transplant recipient (and will therefore be genetically identical to the patient);
2. Hence there is reduced immune incompatibility between donor cells and recipient / lower chance of rejection of transplanted cells;
R! Tissue rejection
3. iPSCs are not derived from human embryos / does not involve isolation and subsequent death of embryos;
4. Hence avoiding ethical issues associated with usage of embryos in scientific research (and scientists are more likely to obtain federal funding and support);
5. It also has fewer ethical issues involved, as genetic makeup of patient's descendants are not altered, unlike in germ-line gene therapy;
6. It is easier to obtain iPSCs in greater numbers compared to embryonic stem cells;
7. iPSCs can be obtained from any differentiated somatic cell while embryonic stem cells can only be obtained from the inner cell mass of the blastocyst;

Yamanaka's research team studied 24 genes expressed by embryonic stem cells in mice to track down genes that can induce pluripotency. They discovered that four genes, notably *Oct4*, *Sox2*, *Klf4* and *c-Myc*, encode transcription factors which could be used to reprogram differentiated cells to form iPSCs. These genes can be introduced into differentiated somatic cells via viral transduction or the introduction of non-integrating plasmids in Fig. 3.2.

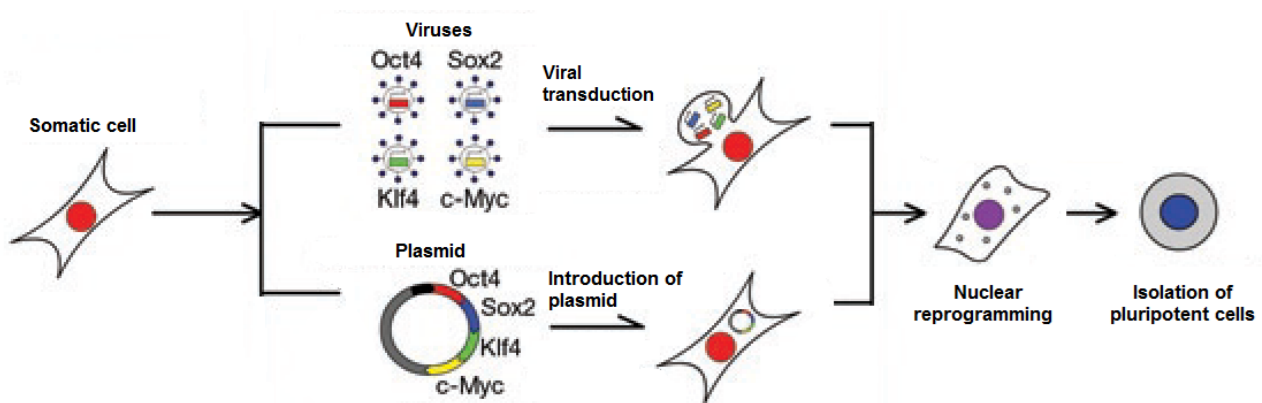


Fig 3.2

(b) (i) Suggest why it is necessary to introduce these four genes into somatic cells in iPSC formation. [1]

These four genes are not expressed / methylated in differentiated (somatic) cells;

R! Genes are not found in somatic cells

(ii) Describe two advantages and one disadvantage of introducing the four genes via non-integrating plasmids over viral transduction using retroviruses. [3]

1. **Advantage #1: Plasmids do not integrate into the genome, hence it does not cause insertional mutagenesis which can facilitate cancer development;**
2. **Advantage #2: Lower risk of stimulating immune response;**

3. Advantage #3: Do not cause disease / lower risk of pathogenicity;
4. Disadvantage #1: Plasmids do not integrate into the genome, hence expression of transgenes is short-lived / cells may only achieve pluripotency temporarily / require multiple rounds of introduction of plasmid;
R! Gene therapy is short-lived (irrelevant context)
5. Disadvantage #2: Lower efficiency of transferring transgenes into cells;

Note: Students must have 2 advantages and 1 disadvantage to get full marks.

(iii) Before iPSCs are stimulated to undergo cellular differentiation, the introduced genes must be removed from the cells. Suggest why this process is necessary. [1]

1. Transcription factors encoded by transgenes allow cells to remain pluripotent which impedes differentiation;
2. To remove the risk of insertional mutagenesis if transgenes are integrated into cells (via retroviral transfection);

[Total: 11m]

4. Planning question

The visking tubing is a semi-permeable membrane which allows smaller molecules such as water and glucose to pass through, but not larger molecules like sucrose and proteins. The tubing is to be supported in a boiling tube as shown in Fig. 4.1.

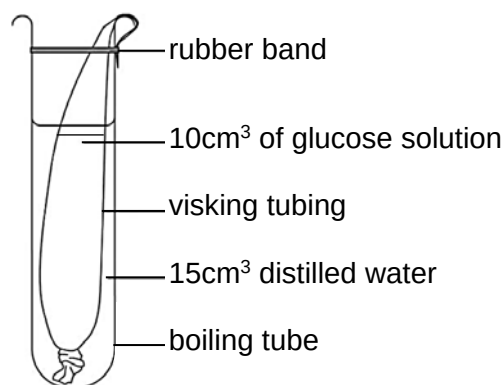


Fig. 4.1

The presence of glucose can be detected using the Benedict's test for reducing sugars.

Using this information and your own knowledge, design an experiment to investigate the effect of glucose concentration on its rate of diffusion across the visking tubing.

You must use:

- 5 pieces of visking tubing (each knotted at one end, and open at the other), soaked in a beaker of distilled water
- 5 boiling tubes and 5 rubber bands
- 2.0 moldm⁻³ glucose solution
- Benedict's solution
- weighing balance and weighing paper
- filter paper and funnel
- distilled water
- thermostatically-controlled water bath

You should select from the following apparatus:

- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, glass rods
- tripod stand, bunsen burner and lighter
- test-tube holder
- stopwatch
- syringes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12m]

Answer Scheme**1. Theoretical consideration or rationale of the plan to justify the practical procedure (1 mark)**

A. How would the independent variable (glucose concentration) affect the dependent variable (rate of diffusion)?

(a) As concentration of glucose in the visking tubing increases, the rate of diffusion of glucose molecules across the membrane increases (over a fixed amount of time);

B. How would the dependent variable (rate of diffusion) be measured?

(b) The rate of diffusion can be measured by weighing the amount of precipitate formed from reacting the solution in the boiling tube with Benedict's solution to test for presence of reducing sugars.

(c) As a result, mass of precipitate (g) formed will increase with an increase in the rate of diffusion.

2 The correct use of technical and scientific terms (throughout the write-up, 1 mark)

(a) down/along concentration gradient,

(b) precipitate

Procedure [max 9 m]

Independent Variable	3. Identification of <u>independent variable</u>: concentration of glucose solutions.;
	4. At least 5 concentration of glucose solutions with regular interval ranging from 0.4, 0.8, 1.2, 1.6, 2.0mol ^{dm} ⁻³ not inclusive of 0.0mol ^{dm} ⁻³
	5. Dilution table to make up different concentrations of glucose solutions (volume of water, volume of glucose solutions, concentration of glucose solutions must be indicated, total volume is the same for all concentrations) Note for markers: Precision does not need to be seen, but if incorrect units penalize.
Dependent Variable	6. Use of filter funnel and filter paper to obtain precipitate
	7. Determination of dependent variable: rate of diffusion measured by measuring the mass of precipitate formed using a weighting balance from reacting glucose with Benedict's solution
	8. Description of Benedict's test: (a) Equal volume of Benedict's solution and test solution (b) Boiling water bath
Constant Variables (Points 9 to 14, max 2)	Constant vol. of glucose solution 10cm ³ (stated in qn, no marks awarded) Constant vol of distilled water 15cm ³ (stated in qn, no marks awarded) 9. Constant vol of solution taken from boiling tube to react with Benedict's solution e.g. 1-5cm³ 10. Constant volume of Benedict's solution used e.g. 1-5cm³ 11. Fixed amount of time for diffusion to occur e.g. 5-30min 12. Fixed amount of time for heating Benedict's solution with sample solution e.g. 1-5min 13. Constant temperature (suitable temperature: 25-35°C) at which diffusion takes place (maintained thermostatically-controlled water bath)

Equilibration	14. Place the (visking tubing in the beaker of distilled water), the glucose solutions and a beaker of distilled water separately in a thermostatically controlled water bath for fixed time e.g. 5 minutes at fixed temperature e.g. 25-30°C.
Control	15. Description of control experiment: (a) Replace glucose solution with 10.0cm³ of distilled water . (as long as present in dilution table, award) (b) Subject the control tube to the same experimental conditions. (c) Ref to rationale of control: This shows that without glucose, red precipitate is not observed as there is no diffusion of glucose across the visking tubing.
Table of results	16. Table of results with clear headings and units . (Concentration of glucose solution/mol dm ⁻³ , Weight of ppt/g, average rate of diffusion of glucose across membrane/g min ⁻¹). Note for markers: ECF. If theory is incorrect, marks can still be awarded.
Repeats and replicates	17. (a) Repeat experiment twice to get replicate readings (at least 3 replicates) (b) Repeat entire experiment twice for repeat readings (at least 2 repeats)
Data Analysis & Graph	18. (a) Plot graphs of average rate of diffusion (g min ⁻¹) against concentration of glucose (mol dm ⁻³) (b) Sketch expected trend, straight line positive gradient graph R: Plateau Note for markers: ECF. If theory is incorrect, marks can still be awarded. However, the y axis of the graph must follow the table seen in 16.
Labelled Diagram	Not required since it is already provided

Points 3-18 max 9 marks

19 Safety Precautions (1 mark)

- Handle the boiling water bath with care as **hot** water can scald / Handle the lit Bunsen burner with care as it can burn/ Handle the test-tubes in the boiling water bath with test-tube holders as the test-tubes can burn
- Wear gloves when handling the Benedict's solution as it is a possible skin irritant.
- Use dry hands when handling the power socket/supply of the thermostatically controlled water bath/ electrical appliances to prevent electrocution.

Procedure

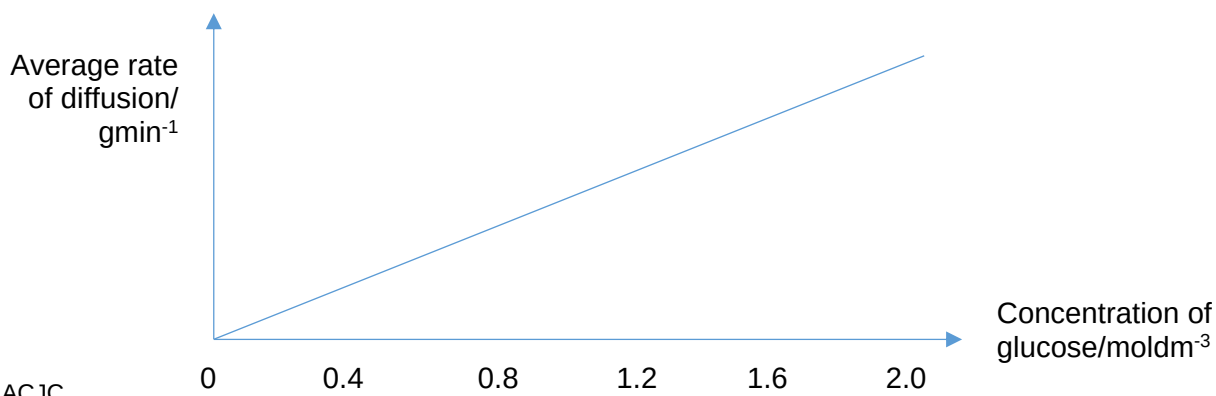
- Label 5 test-tubes 0.4, 0.8, 1.2, 1.6 and 2.0 mol dm⁻³ glucose.
- Using a 10.0 cm³ syringe, dilute the glucose solution, as shown, to obtain the following glucose concentrations:

Conc of glucose solution/mol dm ⁻³	Volume of 2.0 mol dm ⁻³ glucose solution added/cm ³	Volume of distilled water added/cm ³	Total volume/cm ³
0.4	2.0	8.0	10.0
0.8	4.0	6.0	10.0
1.2	6.0	4.0	10.0
1.6	8.0	2.0	10.0
2.0	10.0	0.0	10.0

3. Equilibrate the visking tubing in the beaker of distilled water, the 2.0mol dm^{-3} glucose solution and a beaker of distilled water separately in a thermostatically controlled water bath for 5 minutes at 30°C .
4. Remove a visking tubing from the distilled water and squeeze it to remove as much distilled water as possible.
5. Using a 10.0cm^3 syringe, place 10.0cm^3 of the 2.0mol dm^{-3} glucose solution into the visking tubing and set up the experiment as shown in Fig. 4.1.
6. Start the stopwatch once you place the visking tubing into the boiling tube filled with distilled water and leave the set-up for 10 minutes.
7. After 10min, use a 10.0cm^3 syringe to transfer 2.0cm^3 of the solution in the boiling tube into test-tube labelled 2.0mol dm^{-3} .
8. Using a 10.0cm^3 syringe, add 2.0cm^3 of Benedict's solution to the test-tube.
9. Mix the contents thoroughly and place the test-tubes in a beaker of boiling water for 3 minutes.
10. Remove the test-tube from the boiling water with a test-tube holder.
11. Let it cool before pouring the contents into a funnel lined with filter paper. Use the glass rod to remove all remaining precipitate in the test-tube onto the filter paper.
12. Let the filtrate drip into the sink. When the filtrate stops dripping, remove the filter paper from the funnel.
13. Tare the weighing balance with a clean sheet of weighing paper.
14. Using the glass rod, gently scrap as much of the precipitate from the filter paper onto the weighing boat.
15. Record the weight of the precipitate in a table as shown:

Concentration of glucose solution/ mol dm^{-3}	Weight of precipitate / g				Average rate of diffusion of glucose across membrane/ g min^{-1}
	Replicate 1	Replicate 2	Replicate 3	Average	
0.4					
0.8					
1.2					
1.6					
2.0					

16. Do two more replicates by repeating steps 3 to 15.
17. Repeat steps 3 to 16 for the other glucose concentrations.
18. Repeat the entire experiment twice.
19. To set up the control, repeat steps 3 to 16 using 10.0cm^3 of distilled water instead of glucose. Subject the control tube to the same experimental conditions. This shows that without glucose, red precipitate is not observed as there is no diffusion of glucose across the visking tubing.
20. Calculate the average rate of diffusion of glucose across the visking tubing membrane: average weight of precipitate divided by 10min.
21. Plot a graph to show how average rate of diffusion increases with concentration of the glucose.



Safety Precautions:

- d. Handle the boiling water bath with care as hot water can scald / Handle the lit Bunsen burner with care as it can burn.
- e. Wear gloves when handling the Benedict's solution as it is a possible skin irritant.

5. Free-response Question

Write your answers to this question on the separate paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Describe the steps and advantages of plant tissue culture.

[8]

Techniques (max. 6)

1. Plant tissue culture is performed under aseptic conditions (in a laminar flow hood to prevent bacterial and fungal contamination);
2. A small piece of plant tissue / explant is taken from the meristematic tissue of a stock plant that possess desirable characteristics;
3. Explants are sterilized with bleach / sodium hypochlorite;
4. Explant cells are cultured in a growth medium containing plant growth regulators such as auxin and cytokinin and other nutrients such as sucrose, amino acids, vitamins and inorganic ions (name any 2 nutrients);
5. To allow cells to divide by mitosis to form a callus;
6. Callus is subcultured into separate culture vessels to produce more plantlets;
7. Specific ratio of plant growth regulators such as auxin and cytokinin is then added to induce differentiation of callus into a plantlet;
8. Plantlet is transferred into soil and allowed to grow under controlled conditions in a greenhouse (until plant is hardy enough to survive in the field);

Advantages (max. 4)

9. Production is not affected by weather or environmental conditions;
10. Plants that are difficult to germinate from seeds (e.g., orchids) can be easily produced via tissue culture;
11. Less space is required to grow plants in the lab as compared to the field;
12. Disease-free plants can be produced;
13. Mass production of many copies of genetically identical plants with desirable traits is possible, hence increasing yield of crop;
14. Plant cultures stored in culture flasks are easier to transport (via air freight);

@ 1m, max. 8

- (b) Discuss the benefits and ethical issues related to the use of a named genetically modified animal.

[6]

1. GM salmon;

Beneficial considerations:

2. GM Salmon can produce growth hormones all year round and achieve a faster growth rate to increase yield/ grow to marketable size in half the time required;
3. Make fish farming more environmentally sustainable as reduction in the time required to raise salmon means supply could be increased without proportionately increasing the use of coastal waters;
4. Increase in food conversion rates means that fewer natural resources are required to produce the fish;

Negative ethical implications:

5. Some argue that producing GMOs is tampering with nature as foreign genes are introduced into an organism's genome;
6. Concerns that human genetic modification for enhancement may one day occur;
7. Concerns that genetically modifying salmon may cause suffering or pain or distress / welfare of animals is compromised in the process;
8. There could be unexpected undesirable outcomes / side-effects in the well-being of animals due to unpredictable interaction of the transgenes with its own DNA;
9. Companies (AquaBounty Technologies) have sought to patent GM Salmon that they have developed, and some argue that it reduces animals to the level of objects;
10. Labelling of products on sale to indicate that genetic engineering was involved in their production is not mandatory in some countries;
11. Consumers may thus unknowingly consume products that could be harmful to them / consumption of GM Salmon may produce unintended harmful secondary by-products that lead to potential health risks;
12. GM salmon might upset ecological balance; if the fast-growing salmon escape into the sea, they might affect balance of wild salmon populations as GM salmon may be more aggressive and better at competing with the wild salmon populations for resources;
13. Companies monopolise the technology hence dominating the market because innovations in research are not shared;

(Max. 3)
@ 1m

R: Answers that discuss a named plant organism (awarded 0 marks).
Answers that do not give a named organism, will still be awarded marks for relevant points.

(c) Discuss the goals and benefits of the human genome project.

[6]

Goals

1. Construct a detailed genetic map (i.e. map formed using recombination frequencies and measured in terms of cM) of the entire human genome.;
2. Determine the nucleotide sequences of all 24 human chromosomes (i.e. the physical map of the genome as measured in base pairs) by the year 2005.;
3. Identify all the approximately 20,000-25,000 genes in human DNA.;
4. Improve technology for DNA sequencing and studying the function of DNA on a genomic scale.;
5. Sequence genomes of model organisms (*E. coli*, budding yeast, *C. elegans*, *Drosophila*, and mouse) in order to compare genomes.;
6. Develop bioinformatics support – to (a) create and operate databases for easy access to data and (b) develop and improve tools for data analysis eg. Comparing and interpreting genome information.;
7. Address the ethical, legal and social issues that may arise from the project.;

Max 3

Benefits

8. Genetic testing - Improve diagnosis of disease; predict the risk of future disease in healthy individuals or their progeny
9. Pharmacogenetics - Customised medication and treatment based on your unique genome;
10. Gene therapy - Delivering specific genes into cells to treat disease;
11. Risk assessment of individuals upon exposure to toxic agents - Genetic differences make some people more susceptible and others more resistant to such agents → Assess health damage and risks to individuals caused by exposure to radiation exposure and carcinogens → Evaluate if individuals are suitable to work in those areas with exposure to radiation and carcinogens;
12. Understanding human evolution - Study evolution through germline mutations in lineages/ Compare breakpoints in the evolution of mutations with ages of populations and historical events/ Study migration of different population groups based on female genetic inheritance (mitochondrial DNA) and male genetic inheritance (Y chromosome);
13. DNA forensics - Catch criminals / Identify victims of crimes, disasters / Establish family relationships / Match organ donors and recipients / Identify endangered/protected species / Determine pedigree for seed or livestock / Authenticate consumables such as caviar and wine;

NOTE: each point requires elaboration and can be owtte

Max 3